

miniature data center and relaying information back and forth. They could also act as Wi-Fi hotspots. Sensors could monitor energy consumption and this can cut down on wastage of electricity. It will not be a drastic measure but its a start to help create an energy efficient city. In zones which are prone of natural calamities, appropriate sensors could be installed. There have been instances of sensors being installed in buildings to gather data about energy consumption and share it with other systems to optimise electricity use. This sort of a system could have major implications in disaster warning and management in the event of natural calamities as well. Barcelona has teamed up with Cisco to actually implement the above ideas. The city has a spinal cord like fibre network that is used to connect all of the city's IoT designs. The bus stops are connected to the fibre. The bus stops display bus timings and other related information besides functioning as a Wifi hotspot and charging zone. Mobile users can access information about nearby available parking zones using dedicated apps, thanks to the parking spots also being connected to the fibre network. Barcelona also has a working version of the super-streetlight technology equipped with Wi-Fi and air-monitoring systems. Barcelona has also implemented an IoT based system to monitor garbage levels and optimize garbage collection. There are sensors on bins that inform stations about level of garbage and this data is used to plan efficient routes for garbage collection. Hazardous material detection sen-



A sensor – an integral component of any IoT system